

Semax

Semax — ACTH(4-7)PGP Neuropeptide

NOOTROPIC

NEUROPROTECTIVE

BDNF

FOCUS

COGNITIVE
ENHANCEMENT

WHAT IS IT?

Semax is a synthetic heptapeptide analogue of ACTH(4-7), developed by the Institute of Molecular Genetics (Russian Academy of Sciences) in the 1980s–90s. It is approved in Russia for stroke, TBI, cognitive decline, ADHD, and nervous system disorders. It dramatically increases BDNF (brain-derived neurotrophic factor) and NGF (nerve growth factor) — two of the brain's most important neuroplasticity molecules. Legal and widely used internationally as a research nootropic.

DOSING AT A GLANCE

SC injection	Starting dose	Frequency	Cycle pattern
200–600 mcg	100–200 mcg	1–2x daily	5–10 days on/off

KEY BENEFITS

- Rapidly elevates BDNF and NGF — promotes neurogenesis and synaptic plasticity
- Enhances working memory, focus, and information processing speed
- Neuroprotective — shown effective in stroke, TBI, and ischemia recovery
- Reduces anxiety without sedation (anxiolytic without impairing cognition)
- Improves executive function, learning capacity, and attention
- Anti-fatigue — improves mental endurance and stamina under cognitive load
- Shown to improve ADHD symptoms without stimulant side effects
- Mood enhancement — BDNF is central to antidepressant mechanisms
- Potential neurological disease application: Parkinson's, Alzheimer's (early research)

HOW IT WORKS

Semax is a melanocortin receptor (MCR) agonist that also directly stimulates BDNF and NGF gene expression in the brain. Its proline-glycine-proline (PGP) C-terminal extension makes it resistant to enzymatic degradation compared to native ACTH(4-7). It modulates the dopaminergic and serotonergic systems, and upregulates genes involved in synaptic transmission, neuroplasticity, and immune regulation in the CNS.

PROTOCOLS

- **SC injection (standard):** 200–600 mcg SC, 1–2x daily. Inject abdomen or thigh.
- **Cycle protocol:** 5–10 days on, 5 days off to prevent receptor downregulation.
- **Cognitive enhancement:** 200 mcg SC 30 min before demanding work.
- **Recovery (TBI/stroke):** 600–900 mcg/day SC for 10–14 days under physician guidance.
- **Stack:** Semax (AM) + Selank (PM) — cognitive sharpness + anxiety control without stimulants.

SIDE EFFECTS

- Mild injection site irritation (transient)
- Mild headache at higher doses
- Transient fatigue after dose wears off (BDNF normalization)
- Rarely: mild mood elevation that can overshoot to slight agitation
- No known significant toxicity at therapeutic doses

STORAGE

Reconstituted solution: refrigerate 2–8°C, use within 30 days. Lyophilized: -20°C freezer. Protect from light.

■ ***N-acetyl Semax (NA-Semax) and NA-Semax Amidate are more potent variants — start at lower doses. Refrigeration required — peptide degrades rapidly at room temperature.***

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